The Balance of Payments Accounts

In addition to national income accounts, government economists and statisticians also keep balance of payments accounts, a detailed record of the composition of the current account balance and of the many transactions that finance it.[10] Balance of payments figures are of great interest to the general public, as indicated by the attention that various news media pay to them. But press reports sometimes confuse different measures of international payments flows. Should we be alarmed or cheered by a *Wall Street Journal* headline proclaiming, "U.S. Chalks Up Record Balance of Payments Deficit"? A thorough understanding of balance of payments accounting will help us evaluate the implications of a country's international transactions.

A country's balance of payments accounts keep track of both its payments to and its receipts from foreigners. Any transaction resulting in a receipt from foreigners is entered in the balance of payments accounts as a *credit*. Any transaction resulting in a payment to foreigners is entered as a *debit*. Three types of international transaction are recorded in the balance of payments:

1. Transactions that arise from the export or import of goods or services and therefore enter directly into the current account. When a French consumer imports American blue jeans, for example, the transaction enters the U.S. balance of payments accounts as a credit on the current account.
2. Transactions that arise from the purchase or sale of financial assets. An **asset** is any one of the forms in which wealth can be held, such as money, stocks, factories, or government debt. The **financial account** of the balance of payments records all international purchases or sales of financial assets. When an American company buys a French factory, the transaction enters the U.S. balance of payments as a debit in the financial account. It enters as a debit because the transaction requires a payment from the United States to foreigners. Correspondingly, a U.S. sale of assets to foreigners enters the U.S. financial account as a credit. The difference between a country's purchases and sales of foreign assets is called its *financial account balance*, or its *net financial flows*.
3. Certain other activities resulting in transfers of wealth between countries are recorded in the **capital account**. These international asset movements—which are generally very small for the United States—differ from those recorded in the financial account. For the most part they result from nonmarket activities or represent the acquisition or disposal of nonproduced, nonfinancial, and possibly intangible assets (such as copyrights and trademarks). For example, if the U.S. government forgives $1 billion in debt owed to it by the government of Pakistan, U.S. wealth declines by $1 billion and a $1 billion debit is recorded in the U.S. capital account.

You will find the complexities of the balance of payments accounts less confusing if you keep in mind the following simple rule of double-entry bookkeeping: *Every international transaction automatically enters the balance of payments twice, once as a credit and*

[10]The U.S. Bureau of Economic Analysis (BEA) has changed its balance of payments presentation to conform to prevailing international standards, so our discussion in this chapter differs in some respects from that in editions one through eight of this book. We follow the new methodology described by Kristy L. Howell and Robert E. Yuskavage, "Modernizing and Enhancing BEA's International Economic Accounts: Recent Progress and Future Directions," *Survey of Current Business* (May 2010), pp. 6–20. The BEA completed the full transition to the new system in June 2014. For an update, see Jeffrey R. Bogen, Mai-Chi Hoang, Kristy A. Howell, and Erin M. Whitaker, "Comprehensive Restructuring and Annual Revision of the U.S. International Transactions Accounts," *Survey of Current Business* (July 2014), pp. 1–24.

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once as a debit. This principle of balance of payments accounting holds true because every transaction has two sides: If you buy something from a foreigner, you must pay him in some way, and the foreigner must then somehow spend or store your payment.

Examples of Paired Transactions

Some examples will show how the principle of double-entry bookkeeping operates in practice.

1. Imagine you buy an ink-jet fax machine from the Italian company Olivetti and pay for your purchase with a $1,000 check. Your payment to buy a good (the fax machine) from a foreign resident enters the U.S. current account as a debit. But where is the offsetting balance of payments credit? Olivetti's U.S. salesperson must do something with your check—let's say he deposits it in Olivetti's account at Citibank in New York. In this case, Olivetti has purchased, and Citibank has sold, a U.S. asset—a bank deposit worth $1,000—and the transaction shows up as a $1,000 credit in the U.S. financial account. The transaction creates the following two offsetting bookkeeping entries in the U.S. balance of payments:

	Credit	Debit
Fax machine purchase (Current account, U.S. good import)		$1,000
Sale of bank deposit by Citibank (Financial account, U.S. asset sale)	$1,000	

2. As another example, suppose that during your travels in France, you pay $200 for a fine dinner at the Restaurant de l'Escargot d'Or. Lacking cash, you place the charge on your Visa credit card. Your payment, which is a tourist expenditure, will be counted as a service import for the United States, and therefore as a current account debit. Where is the offsetting credit? Your signature on the Visa slip entitles the restaurant to receive $200 (actually, its local currency equivalent) from First Card, the company that issued your Visa card. It is therefore an asset, a claim on a future payment from First Card. So when you pay for your meal abroad with your credit card, you are selling an asset to France and generating a $200 credit in the U.S. financial account. The pattern of offsetting debits and credits in this case is:

	Credit	Debit
Meal purchase (Current account, U.S. service import)		$200
Sale of claim on First Card (Financial account, U.S. asset sale)	$200	

3. Imagine next that your Uncle Sid from Los Angeles buys a newly issued share of stock in the U.K. oil giant British Petroleum (BP). He places his order with his U.S. stockbroker, Go-for-Broke, Inc., paying $95 with funds from his Go-for-Broke money market account. BP, in turn, deposits the $95 Sid has paid into its own U.S. bank account at Second Bank of Chicago. Uncle Sid's acquisition of the stock creates a $95 debit in the U.S. financial account (he has purchased an

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asset from a foreign resident, BP), while BP's $95 deposit at its Chicago bank is the offsetting financial account credit (BP has expanded its U.S. asset holdings). The mirror-image effects on the U.S. balance of payments therefore both appear in the financial account:

	Credit	Debit
Uncle Sid's purchase of a share of BP (Financial account, U.S. asset purchase)		$95
BP's deposit of Uncle Sid's payment at Second Bank of Chicago (Financial account, U.S. asset sale)	$95	

4. Finally, let's consider how the U.S. balance of payments accounts are affected when U.S. banks forgive (that is, announce that they will simply forget about) $5,000 in debt owed to them by the government of the imaginary country of Bygonia. In this case, the United States makes a $5,000 capital transfer to Bygonia, which appears as a $5,000 debit entry in the capital account. The associated credit is in the financial account, in the form of a $5,000 reduction in U.S. assets held abroad (a negative "acquisition" of foreign assets, and therefore a balance of payments credit):

	Credit	Debit
U.S. Banks' debt forgiveness (Capital account, U.S. transfer payment)		$5,000
Reduction in banks' claims on Bygonia (Financial account, U.S. asset sale)	$5,000	

These examples show that many circumstances can affect the way a transaction generates its offsetting balance of payments entry. We can never predict with certainty where the flip side of a particular transaction will show up, but we can be sure that it will show up somewhere.

The Fundamental Balance of Payments Identity

Because any international transaction automatically gives rise to offsetting credit and debit entries in the balance of payments, the sum of the current account balance and the capital account balance automatically equals the financial account balance:

$$\text{Current account} + \text{capital account} = \text{Financial account}. \qquad (2\text{-}3)$$

In examples 1, 2, and 4 previously, current or capital account entries have offsetting counterparts in the financial account, while in example 3, two financial account entries offset each other.

You can understand this identity another way. Recall the relationship linking the current account to international lending and borrowing. Because the sum of the current and capital accounts is the total change in a country's net foreign assets (including, through the capital account, nonmarket asset transfers), that sum necessarily equals the difference between a country's purchases of assets from foreigners and its sales of assets to them—that is, the financial account balance (also called net financial flows).

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We now turn to a more detailed description of the balance of payments accounts, using as an example the U.S. accounts for 2015.

The Current Account, Once Again

As you have learned, the current account balance measures a country's net exports of goods and services. Table 2-2 shows that U.S. exports (on the credit side) were $3,044.08 billion in 2015, while U.S. imports (on the debit side) were $3,362.06 billion.

The balance of payments accounts divide exports and imports into three finer categories. The first is *goods* trade, that is, exports or imports of merchandise. The second category, *services*, includes items such as payments for legal assistance, tourists' expenditures, and shipping fees. The final category, *income*, is made up mostly of international interest and dividend payments and the earnings of domestically owned firms

Pearson MyLab Economics Real-time data

TABLE 2-2 U.S. Balance of Payments Accounts for 2015 (billions of dollars)

Current Account	
(1) Exports	
Of which:	3,044.08
Goods	
Services	1,510.30
Income receipts (primary income)	750.86
(2) Imports	782.92
Of which:	3,362.06
Goods	
Services	2,272.87
Income payments (primary income)	488.66
(3) Net unilateral transfers (secondary income)	600.53
Balance on current account	−144.99
[(1) − (2) + (3)]	−462.97
Capital Account	
(4)	
Financial Account	−0.04
(5) Net U.S. acquisition of financial assets, excluding financial derivatives	225.40
Of which:	
Official reserve assets	
Other assets	−6.29
(6) Net U.S. incurrence of liabilities, excluding financial derivatives	231.69
Of which:	395.23
Official reserve assets	
Other assets	−98.10
(7) Financial derivatives, net	493.33
Net financial flows	−25.39
[(5) − (6) + (7)]	−195.23
Statistical Discrepancy	
[Net financial flows less sum of current and capital accounts]	267.78

Source: U.S. Department of Commerce, Bureau of Economic Analysis, June 16, 2016, release. Totals may differ from sums because of rounding.

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operating abroad. If you own a share of a German firm's stock and receive a dividend payment of $5, that payment shows up in the accounts as a U.S. investment income receipt of $5. Wages that workers earn abroad can also enter the income account.

We include income on foreign investments in the current account because that income really is compensation for the *services* provided by foreign investments. This idea, as we saw earlier, is behind the distinction between GNP and GDP. When a U.S. corporation builds a plant in Canada, for instance, the productive services the plant generates are viewed as a service export from the United States to Canada equal in value to the profits the plant yields for its American owner. To be consistent, we must be sure to include these profits in American GNP and not in Canadian GNP. Remember, the definition of GNP refers to goods and services generated by a country's factors of production, but it does *not* specify that those factors must work within the borders of the country that owns them. The earnings of capital and labor working abroad are referred to as "primary income."

Before calculating the current account, we must include one additional type of international transaction that we have largely ignored until now. In discussing the relationship between GNP and national income, we defined unilateral transfers between countries as international gifts, that is, payments that do not correspond to the purchase of any good, service, or asset. Such payments are referred to as "secondary income." Net unilateral transfers are considered part of the current account as well as a part of national income, and the identity $Y = C + I + G + CA$ holds exactly if Y is interpreted as GNP *plus* net transfers. In 2015, the U.S. balance of unilateral transfers was −$144.99 billion.

The table shows a 2015 current account balance of $3,044.08 billion − $3,362.06 billion − $144.99 billion = −$462.97 billion, a deficit.

The negative sign means that current payments to foreigners exceeded current receipts and that U.S. residents used more output than they produced. Since these current account transactions were paid for in some way, we know that this $462.97 billion net debit entry must be offset by a net $462.97 billion credit elsewhere in the balance of payments.

The Capital Account

The capital account entry in Table 2-2 shows that in 2015, the United States payed about $40 million in net capital transfers. The net balance of −$40 million is a balance of payments debit. After we add it to the payments deficit implied by the current account, we find that the United States' need to cover its excess payments to foreigners is increased very slightly, from $462.97 billion to $463.01 billion. Because an excess of national spending over income must be covered by net borrowing from foreigners, this negative current plus capital account balance must be matched by an equal negative balance of net financial flows, representing the net liabilities the United States incurred to foreigners in 2015 in order to fund its deficit.

The Financial Account

While the current account is the difference between sales of goods and services to foreigners and purchases of goods and services from them, the financial account measures the difference between acquisitions of assets from foreigners and the buildup of liabilities to them. When the United States borrows $1 from foreigners, it is selling them an asset—a promise that they will be repaid $1, with interest, in the future. Likewise, when the United States lends abroad, it acquires an asset: the right to claim future repayment from foreigners.

To cover its 2015 current plus capital account deficit of $463.01 billion, the United States needed to borrow from foreigners (or otherwise sell assets to them) in the net

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amount of $463.01 billion. We can look again at Table 2-2 to see exactly how this net sale of assets to foreigners came about.

The table records separately U.S. acquisitions of foreign financial assets (which are balance of payments debits, because the United States must pay foreigners for those assets) and increases in foreign claims on residents of the United States (which are balance of payments credits, because the United States receives payments when it sells assets overseas).

These data on increases in U.S. asset holdings abroad and foreign holdings of U.S. assets do not include holdings of *financial derivatives*, which are a class of assets that are more complicated than ordinary stocks and bonds, but have values that can depend on stock and bond values. (We will describe some specific derivative securities in the next chapter.) Starting in 2006, the U.S. Department of Commerce was able to assemble data on *net* cross-border derivative flows for the United States (U.S. net purchases of foreign-issued derivatives less foreign net purchases of U.S.-issued derivatives). Derivatives transactions enter the balance of payments accounts in the same way as do other international asset transactions.

According to Table 2-2, U.S.-owned assets abroad (other than derivatives) increased (on a net basis) by $225.40 billion in 2015. The figure is "on a net basis" because some U.S. residents bought foreign assets while others sold foreign assets they already owned, the difference between U.S. gross purchases and sales of foreign assets being $225.40 billion. In the same year (again on a net basis), the United States incurred new liabilities to foreigners equal to $395.23 billion. Some U.S. residents undoubtedly repaid foreign debts, but new borrowing from foreigners exceeded these repayments by $395.23 billion. The balance of U.S. purchases and sales of financial derivatives was −$25.39 billion: The United States acquired derivative claims on foreigners that were lower in value than the derivative claims on the U.S. that foreigners acquired. We calculate the balance on financial account (net financial flows) as $225.40 billion − $395.23 billion − $25.39 billion = −$195.23 billion. The negative value for net financial flows means that in 2015, the United States increased its net liability to foreigners (liabilities minus assets) by $195.23 billion.

Statistical Discrepancy

We come out with net financial flows of −$195.23 billion rather than the −$463.01 billion that we'd expected as a result of adding up the current and capital account balances. According to our data on trade and financial flows, the United States incurred $267.78 billion less in foreign debt than it actually needed to fund its current plus capital account deficit—because (−$195.23 billion) − (−$463.01 billion) = $267.78 billion. If every balance of payments credit automatically generates an equal counterpart debit and vice versa, how is this difference possible? The reason is that information about the offsetting debit and credit items associated with a given transaction may be collected from different sources. For example, the import debit that a shipment of DVD players from Japan generates may come from a U.S. customs inspector's report and the corresponding financial account credit from a report by the U.S. bank in which the check paying for the DVD players is deposited. Because data from different sources may differ in coverage, accuracy, and timing, the balance of payments accounts seldom balance in practice as they must in theory. Account keepers force the two sides to balance by adding to the accounts a *statistical discrepancy* item. For 2015, unrecorded (or misrecorded) international transactions generated a balancing accounting credit of $267.78 billion—the difference between the recorded net financial flows and the sum of the recorded current and capital accounts.

We have no way of knowing exactly how to allocate this discrepancy among the current, capital, and financial accounts. (If we did, it wouldn't be a discrepancy!) The financial account is the most likely culprit, since it is notoriously difficult to keep track of the complicated financial trades between residents of different countries. But we cannot conclude that net financial flows were $267.78 billion lower than recorded because the current account is also highly suspect. Balance of payments accountants consider merchandise trade data relatively reliable, but data on services are not. Service transactions such as sales of financial advice and computer programming assistance may escape detection. Accurate measurement of international interest and dividend receipts can be particularly difficult.

Official Reserve Transactions

Although there are many types of financial account transactions, one type is important enough to merit separate discussion. This type of transaction is the purchase or sale of official reserve assets by central banks.

An economy's **central bank** is the institution responsible for managing the supply of money. In the United States, the central bank is the Federal Reserve System. **Official international reserves** are foreign assets held by central banks as a cushion against national economic misfortune. At one time, official reserves consisted largely of gold, but today, central banks' reserves include substantial foreign financial assets, particularly U.S. dollar assets such as Treasury bills. The Federal Reserve itself holds only a small level of official reserve assets other than gold; its own holdings of U.S. dollar assets are not considered international reserves.

Central banks often buy or sell international reserves in private asset markets to affect macroeconomic conditions in their economies. Official transactions of this type are called **official foreign exchange intervention.** One reason why foreign exchange intervention can alter macroeconomic conditions is that it is a way for the central bank to inject money into the economy or withdraw it from circulation. We will have much more to say later about the causes and consequences of foreign exchange intervention.

Government agencies other than central banks may hold foreign reserves and intervene officially in exchange markets. The U.S. Treasury, for example, operates an Exchange Stabilization Fund that at times has played an active role in market trading. Because the operations of such agencies usually have no noticeable impact on the money supply, however, we will simplify our discussion by speaking (when it is not too misleading) as if the central bank alone holds foreign reserves and intervenes.

When a central bank purchases or sells a foreign asset, the transaction appears in its country's financial account just as if the same transaction had been carried out by a private citizen. A transaction in which the central bank of Japan (the Bank of Japan) acquires dollar assets might occur as follows: A U.S. auto dealer imports a Nissan sedan from Japan and pays the auto company with a check for $20,000. Nissan does not want to invest the money in dollar assets, but it so happens that the Bank of Japan is willing to give Nissan Japanese money in exchange for the $20,000 check. The Bank of Japan's international reserves rise by $20,000 as a result of the deal. Because the Bank of Japan's dollar reserves are part of total Japanese assets held in the United States, the latter rise by $20,000. This transaction therefore results in a $20,000 credit in the U.S. financial account, the other side of the $20,000 debit in the U.S. current account due to the import of the car.[11]

[11] To test your understanding, see if you can explain why the same sequence of actions causes a $20,000 improvement in Japan's current account and a $20,000 increase in its net financial flows.

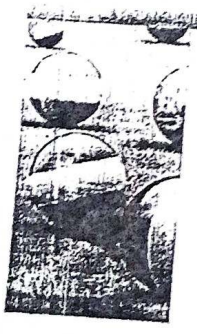
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Table 2-2 shows that in 2015, U.S. official reserve assets fell by $6.29 billion. As shown in the table, foreign central banks sold $98.10 billion of the U.S. reserves they previously held. The net increase in U.S. official reserves *less* the increase in foreign official reserve claims on the United States is the level of net central bank financial flows, which stood at −$6.29 billion − (−$98.10) = $91.81 billion in 2015.

You can think of this $91.81 billion net central bank financial flow as measuring the degree to which monetary authorities in the United States and abroad generated additional net U.S. claims on foreigners—which help to finance the U.S. current account deficit when the number is negative, but increase the need for private foreign financing when it is positive. In the example above, the Bank of Japan, by acquiring a $20,000 U.S. bank deposit, indirectly finances an American import of a $20,000 Japanese car. The level of net central bank financial flows is called the **official settlements balance** or (in less formal usage) the **balance of payments.** This balance is the sum of the current account and capital account balances, less the nonreserve portion of the financial account balance, and it indicates the role of central banks' official reserve transactions in offsetting the current account balance. Thus, the U.S. balance of payments in 2015 was $91.81 billion.

The balance of payments played an important historical role as a measure of disequilibrium in international payments, and for many countries it still plays this role. A negative balance of payments (a deficit) may signal a crisis, for it means that a country is running down its international reserve assets or incurring debts to foreign monetary authorities. If a country faces the risk of being suddenly cut off from foreign loans, it will want to maintain a "war chest" of international reserves as a precaution. Many developing countries, in particular, are in this position (see Chapter 11).

Like any summary measure, however, the balance of payments must be interpreted with caution. To return to our running example, the Bank of Japan's decision to expand its U.S. bank deposit holdings by $20,000 swells the measured U.S. balance of payments deficit by the same amount. Suppose the Bank of Japan instead places its $20,000 with Barclays Bank in London, which in turn deposits the money with Citibank in New York. The United States incurs an extra $20,000 in liabilities to *private* foreigners in this case, and the U.S. balance of payments deficit does not rise. But this "improvement" in the balance of payments is of little economic importance: It makes no real difference to the United States whether it borrows the Bank of Japan's money directly or through a London bank.



CASE STUDY The Assets and Liabilities of the World's Biggest Debtor

We saw earlier that the current account balance measures the flow of new net claims on foreign wealth that a country acquires by exporting more goods and services than it imports. This flow is not, however, the only important factor that causes a country's net foreign wealth to change. In addition, changes in the market price of wealth previously acquired can alter a country's net foreign wealth. When Japan's stock market lost three-quarters of its value over the 1990s, for example, American and European owners of Japanese shares saw the value of their claims on Japan plummet, and Japan's net *foreign* wealth increased as a result. Exchange

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rate changes have a similar effect. When the dollar depreciates against foreign currencies, for example, foreigners who hold dollar assets see their wealth fall when measured in their home currencies.

The Bureau of Economic Analysis (BEA) of the U.S. Department of Commerce, which oversees the vast job of data collection behind the U.S. national income and balance of payments statistics, reports annual estimates of the net international investment position of the United States—the country's foreign assets less its foreign liabilities. Because asset price and exchange rate changes alter the dollar values of foreign assets and liabilities alike, the BEA must adjust the values of existing claims to reflect such capital gains and losses in order to estimate U.S. net foreign wealth. These estimates show that at the end of 2015, the United States had a *negative* net foreign wealth position far greater than that of any other country.

Until 1991, foreign direct investments such as foreign factories owned by U.S. corporations were valued at their historical, that is, original, purchase prices. Now the BEA uses two different methods to place current values on foreign direct investments: the *current cost* method, which values direct investments at the cost of buying them today, and the *market value* method, which is meant to measure the price at which the investments could be sold. These methods can lead to different valuations because the cost of replacing a particular direct investment and the price it would command if sold on the market may be hard to measure. (The net foreign wealth data graphed in Figure 2-2 are current cost estimates, which are believed to be more accurate.)

Table 2-3 reproduces the BEA's account of how it made its valuation adjustments to find the U.S. net IIP at the end of 2015. This "headline" estimate values direct investments at current cost. Starting with its estimate of 2014 net foreign wealth (−$7,046.1 billion), the BEA added the amount of the 2015 U.S. net financial flow of −$195.2 billion—recall the figure reported in Table 2-2. Then the BEA adjusted the values of previously held assets and liabilities for various changes in their dollar prices. As a result of these valuation changes, U.S. net foreign wealth fell by an amount greater than the $195.2 billion in new net borrowing from foreigners—in fact, U.S. net foreign wealth declined by $234.5 billion. The BEA's 2015 estimate of U.S. net foreign wealth, therefore, was −$7,280.6 billion.

This debt is larger than the total foreign debt owed by all the Central and Eastern European countries, which was about $822 billion in 2014. To put these figures in perspective, however, it is important to realize that the U.S. net foreign debt amounted to about 40 percent of its GDP, while the foreign liability of Hungary, Poland, Romania, and the other Central and Eastern European debtors was about 55 percent of their collective GDP! Thus, the U.S. external debt represents a lower domestic income drain.

Changes in exchange rates and securities prices have the potential to change the U.S. net foreign debt sharply, however, because the *gross* foreign assets and liabilities of the United States have become so large in recent years. Figure 2-3 illustrates this dramatic trend. In 1976, U.S. foreign assets stood at only 20 percent of U.S. GDP and liabilities at 15 percent (making the United States a net foreign creditor in the amount of roughly 5 percent of its GDP). In 2015, however, the country's foreign assets amounted to roughly 130 percent of GDP and its liabilities to roughly 171 percent. The tremendous growth in these stocks of wealth reflects the rapid globalization of financial markets in the late 20th century, a phenomenon we will discuss further in Chapter 9.

TABLE 2-3 Change in the Yearend U.S. Net International Investment Position, 2014 (Billions of Dollars)

Line	Type of investment	Yearend position, 2014[r]	Change in position in 2015						Yearend position, 2015[p]
			Total	Attributable to					
				Financial-account transactions	Other changes in position				
					Total	Price changes	Exchange-rate changes[1]	Changes in volume and valuation[2]	
1	U.S. net international investment position (line 4 less line 35)	−7,046.1	−234.5	−195.2	−39.3	(4)	(4)	(4)	−7,280.6
2	Net international investment position excluding financial derivatives (line 5 less line 36)	−7,131.7	−206.2	−169.8	−36.4	781.9	−1,031.5	213.1	−7,337.9
3	Financial derivatives other than reserves, net (line 6 less line 37) 3	85.5	−28.3	−25.4	−2.9	(4)	(4)	(4)	57.2
4	U.S. assets	24,717.5	−1,376.8	(3)	(3)	(3)	(3)	(3)	23,340.8
5	Assets excluding financial derivatives (sum of lines 7, 10, 21, and 26)	21,503.4	−558.0	225.4	−783.4	220.4	−1,141.5	137.7	20,945.4
6	Financial derivatives other than reserves, gross positive fair value (line 15)	3,214.1	−818.8	(3)	(3)	(3)	(3)	(3)	2,395.4
	By functional category:								
7	Direct investment at market value	7,133.1	−154.8	348.6	−503.4	−64.7	−419.3	10.5	6,978.3
8	Equity	6,045.1	−234.0	316.3	−550.3	−64.7	−419.3	−16.3	5,811.1
9	Debt instruments	1,088.1	79.2	32.3	46.9	.....	.....	46.9	1,167.3
10	Portfolio investment	9,704.3	−98.1	154.0	−252.1	333.8	−641.1	56.0	9,606.2
11	Equity and investment fund shares	6,770.6	57.6	202.6	−145.0	357.9	−562.9	60.1	6,828.2
12	Debt securities	2,933.6	−155.7	−48.6	−107.1	−24.0	−78.4	−4.1	2,777.9
13	Short term	447.2	39.1	42.3	−3.4	.....	−3.4	0.0	486.3
14	Long term	2,486.4	−194.7	−91.1	−103.6	−24.0	−75.0	−4.1	2,291.7
15	Financial derivatives other than reserves, gross positive fair value	3,214.1	−818.8	(3)	(3)	(3)	(3)	(3)	2,395.4
16	Over-the-counter contracts	3,144.0	−797.3	(3)	(3)	(3)	(3)	(3)	2,346.7
17	Single-currency interest rate contracts	2,451.1	−643.7	(3)	(3)	(3)	(3)	(3)	1,807.4
18	Foreign exchange contracts	413.4	−73.2	(3)	(3)	(3)	(3)	(3)	342.3
19	Other contracts	277.5	−80.4	(3)	(3)	(3)	(3)	(3)	197.1
20	Exchange-traded contracts	70.1	−21.5	(3)	(3)	(3)	(3)	(3)	48.6
21	Other investment	4,231.8	−254.5	270.9	−16.4	.....	−54.7	71.1	3,977.3
22	Currency and deposits	1,785.5	−156.9	194.4	−37.5	.....	−30.6	−68.0	1,628.6
23	Loans	2,399.2	−95.3	74.8	−20.5	.....	−23.6	3.1	2,304.0
24	Insurance technical reserves	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
25	Trade credit and advances	47.0	−2.3	−1.7	−0.6	.....	−0.6	0.0	44.7
26	Reserve assets	434.3	−50.7	−6.3	−44.4	−38.2	−6.2	0.0	383.6
27	Monetary gold	315.4	−38.2	0.0	−38.2	−38.2	.....	0.0	277.2
28	Special drawing rights	51.9	2.3	(*)	2.3	.....	2.3	0.0	49.7
29	Reserve position in the International Monetary Fund	25.2	−7.6	−6.5	−1.1	.....	−1.1	0.0	17.6
30	Other reserve assets	41.8	2.7	0.2	2.5	0.0	2.5	0.0	39.1
31	Currency and deposits	19.0	−1.4	(*)	−1.4	.....	−1.3	−0.1	17.6
32	Securities	22.8	−1.2	0.2	−1.4	0.0	−1.3	−0.1	21.6
33	Financial derivatives	.....	.....	.....	.....	.....	.....	.....	.....
34	Other claims	0.0	0.0	0.0	0.0	.....	0.0	0.0	0.0
35	U.S. liabilities	31,763.7	−1,142.3	(3)	(3)	(3)	(3)	(3)	30,621.4
36	Liabilities excluding financial derivatives (sum of lines 38, 41, and 56)	28,635.1	−351.8	395.2	−747.0	−561.0	90.0	−96.0	28,283.3
37	Financial derivatives other than reserves, gross negative fair value (line 50)	3,128.6	−790.5	(3)	(3)	(3)	(3)	(3)	2,338.1
	By functional category:								
38	Direct investment at market value	6,350.1	193.8	379.4	−185.7	−160.3	.....	−25.3	6,543.8
39	Equity	4,884.1	95.2	301.1	−205.9	−160.3	.....	−45.6	4,979.3
40	Debt instruments	1,466.0	98.6	78.3	20.2	.....	.....	20.2	1,564.5
41	Portfolio investment	10,919.8	242.8	250.9	−493.7	−400.7	57.0	−36.0	10,677.0
42	Equity and investment fund shares	6,642.5	−423.6	178.3	−245.4	−187.8	.....	−57.5	6,218.9
43	Debt securities	10,277.3	180.8	429.2	−248.4	−212.9	−57.0	21.5	10,458.1
44	Short term	911.8	43.4	45.8	−2.4	.....	−2.4	0.0	955.2
45	Treasury bills and certificates	671.6	53.1	53.1	0.0	.....	.....	0.0	724.7
46	Other short-term securities	240.2	−9.7	−7.3	−2.4	.....	−2.4	0.0	230.5
47	Long term	9,365.5	137.5	383.4	−246.0	−212.9	−54.6	21.5	9,503.0
48	Treasury bonds and notes	5,484.4	61.0	4.8	−56.3	−56.3	.....	0.0	5,423.4
49	Other long-term securities	3,881.1	198.5	388.2	−189.7	−156.6	−54.0	21.5	4,079.6
50	Financial derivatives other than reserves, gross negative fair value	3,128.6	−790.5	(3)	(3)	(3)	(3)	(3)	2,338.1
51	Over-the-counter contracts	3,062.6	−771.5	(3)	(3)	(3)	(3)	(3)	2,291.1
52	Single-currency interest rate contracts	2,398.8	−643.4	(3)	(3)	(3)	(3)	(3)	1,755.4
53	Foreign exchange contracts	393.6	−49.6	(3)	(3)	(3)	(3)	(3)	344.0
54	Other contracts	270.2	−78.5	(3)	(3)	(3)	(3)	(3)	191.7
55	Exchange-traded contracts	66.0	−19.0	(3)	(3)	(3)	(3)	(3)	47.0
56	Other investment	5,365.2	−302.7	−235.1	−67.6	.....	−33.0	−34.6	5,062.5

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Line	Type of investment	Yearend position, 2014 [r]	Change in position in 2015						Yearend position, 2015 [r]
			Total	Attributable to:					
				Financial-account transactions	Other changes in position				
					Total	Price changes	Exchange-rate changes [1]	Changes in volume and valuation n.i.e. [2]	
57	Currency and deposits	2,886.7	27.5	33.4	−5.9	. . . .	−10.8	4.9	2,914.3
58	Loans	2,265.6	−342.0	−282.7	−59.3	. . . .	−19.7	−39.5	1,923.6
59	Insurance technical reserves	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
60	Trade credit and advances	161.7	13.9	14.2	−0.3	. . . .	−0.3	0.0	175.6
61	Special drawing rights allocations	51.2	−2.2	0.0	−2.2	. . . .	−2.2	0.0	48.9

r Revised n.a. Not available . . . . Not applicable (*) Value between zero and +/− $50 million

1. Represents gains or losses on foreign-currency-denominated assets and liabilities due to their revaluation at current exchange rates.
2. Includes changes due to year-to-year shifts in the composition of reporting panels and to the incorporation of more comprehensive survey results. Also includes capital gains and losses of direct investment affiliates and changes in positions that cannot be allocated to financial transactions, price changes, or exchange-rate changes.
3. Financial transactions and other changes in financial derivatives positions are available only on a net basis, which is shown on line 3; they are not separately available for gross positive fair values and gross negative fair values of financial derivatives.
4. Data are not separately available for price changes, exchange-rate changes, and changes in volume and valuation not included elsewhere.

Note: Details may not add to totals because of rounding.

Source: U.S. Bureau of Economic Analysis.

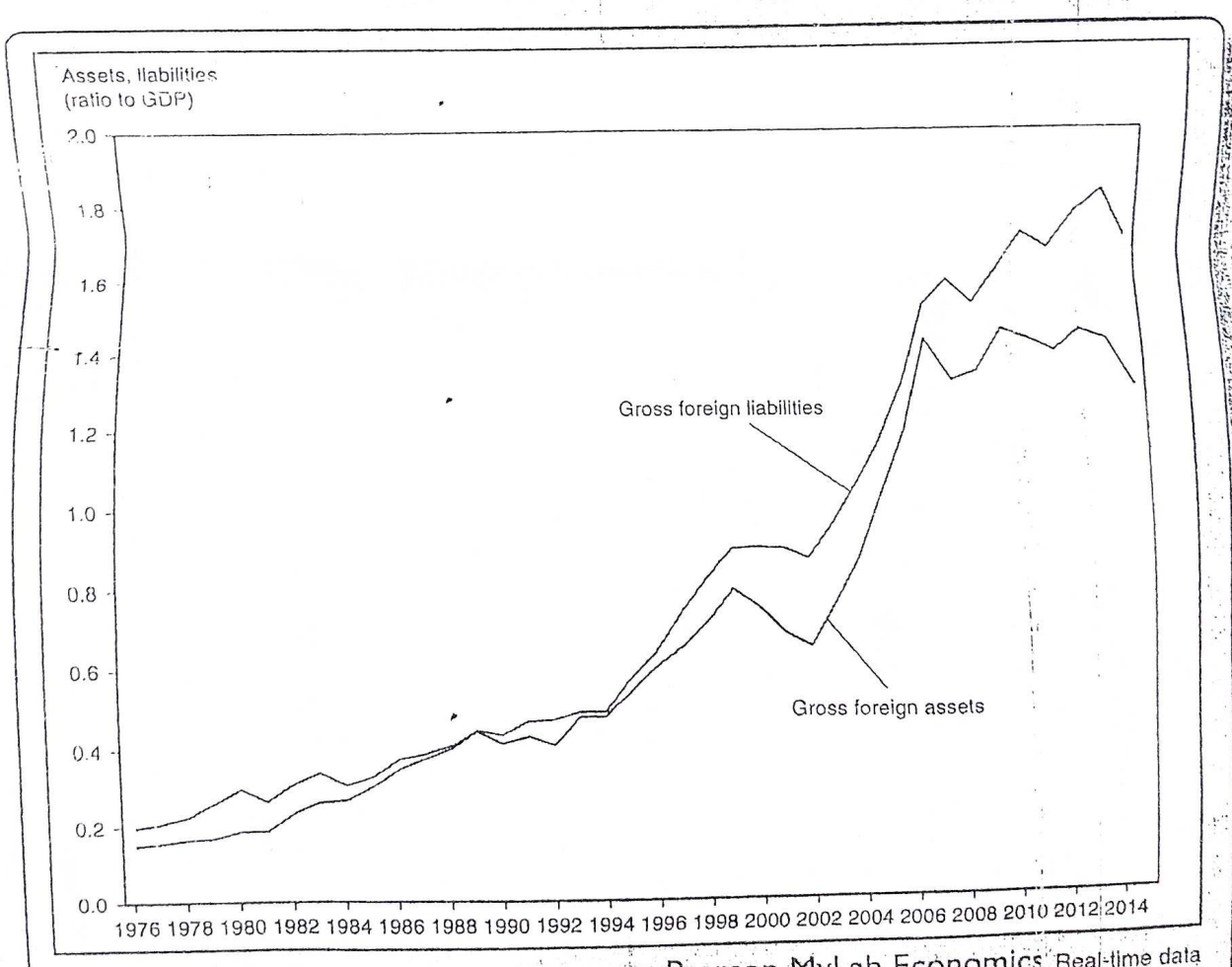


FIGURE 2-3

Gross Foreign Assets and Liabilities, 1976–2015

Since 1976, both the foreign assets and the liabilities of the United States have increased sharply. But liabilities have risen more quickly, leaving the United States with a substantial net foreign debt.

Source: U.S. Department of Commerce, Bureau of Economic Analysis, June 2016.

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Think about how gross wealth positions of this magnitude amplify the effects of exchange rate changes, however. Suppose 70 percent of U.S. foreign assets are denominated in foreign currencies, but all U.S. liabilities to foreigners are denominated in dollars (these are approximately the correct numbers). Because in 2015 U.S. GDP was around $18 trillion, a 10 percent depreciation of the dollar would leave U.S. liabilities unchanged but would increase U.S. assets (measured in dollars) by $0.1 \times 0.7 \times 1.30 = 9.1$ percent of GDP, or about $1.6 trillion. This number is approximately 3.4 times the U.S. current account deficit of 2015! Indeed, due to sharp movements in exchange rates and stock prices, the U.S. economy lost about $800 billion in this way between 2007 and 2008 and gained a comparable amount between 2008 and 2009 (see Figure 2-2). The corresponding redistribution of wealth between foreigners and the United States would have been much smaller back in 1976.

Does this possibility mean that policy makers should ignore their countries' current accounts and instead try to manipulate currency values to prevent large buildups of net foreign debt? That would be a perilous strategy because, as we will see in the next chapter, expectations of future exchange rates are central to market participants' behavior. Systematic government attempts to reduce foreign investors' wealth through exchange rate changes would sharply reduce foreigners' demand for domestic currency assets, thus decreasing or eliminating any wealth benefit from depreciating the home currency.

SUMMARY

1. International *macroeconomics* is concerned with the full employment of scarce economic resources and price level stability throughout the world economy. Because they reflect national expenditure patterns and their international repercussions, the *national income accounts* and the *balance of payments accounts* are essential tools for studying the macroeconomics of open, interdependent economies.
2. A country's *gross national product* (GNP) is equal to the income received by its factors of production. The national income accounts divide national income according to the types of spending that generate it: *consumption*, *investment*, *government purchases*, and the *current account balance*. *Gross domestic product* (GDP), equal to GNP less net receipts of factor income from abroad, measures the output produced within a country's territorial borders.
3. In an economy closed to international trade, GNP must be consumed, invested, or purchased by the government. By using current output to build plant, equipment, and inventories, investment transforms present output into future output. For a closed economy, investment is the only way to save in the aggregate, so the sum of the saving carried out by the private and public sectors, *national saving*, must equal investment.
4. In an open economy, GNP equals the sum of consumption, investment, government purchases, and net exports of goods and services. Trade does not have to be balanced if the economy can borrow from and lend to the rest of the world. The difference between the economy's exports and imports, the current account balance, equals the difference between the economy's output and its total use of goods and services.
5. The current account also equals the country's net lending to foreigners. Unlike a closed economy, an open economy can save through domestic *and* foreign

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investments. National saving therefore equals domestic investment plus the current account balance. The current account is closely related to the change in the *net international investment position*, though usually not equal to that change because of fluctuations in asset values not recorded in the national income and product accounts.

6. Balance of payments accounts provide a detailed picture of the composition and financing of the current account. All transactions between a country and the rest of the world are recorded in the country's balance of payments accounts. The accounts are based on the convention that any transaction resulting in a payment to foreigners is entered as a debit while any transaction resulting in a receipt from foreigners is entered as a credit.
7. Transactions involving goods and services appear in the current account of the balance of payments, while international sales or purchases of *assets* appear in the *financial account*. The *capital account* records mainly nonmarket asset transfers and tends to be small for the United States. The sum of the current and capital account balances must equal the financial account balance (net financial flows). This feature of the accounts reflects the fact that discrepancies between export earnings and import expenditures must be matched by a promise to repay the difference, usually with interest, in the future.
8. International asset transactions carried out by *central banks* are included in the financial account. Any central bank transaction in private markets for foreign currency assets is called *official foreign exchange intervention*. One reason intervention is important is that central banks use it as a way to change the amount of money in circulation. A country has a deficit in its *balance of payments* when it is running down its *official international reserves* or borrowing from foreign central banks; it has a surplus in the opposite case.

KEY TERMS

asset, p. 44
balance of payments accounting, p. 33
capital account, p. 44
central bank, p. 50
consumption, p. 36
current account balance, p. 38
financial account, p. 44
government budget deficit, p. 42
government purchases, p. 37
gross domestic product (GDP), p. 35
gross national product (GNP), p. 33
investment, p. 36
macroeconomics, p. 31
microeconomics, p. 31
national income, p. 34
national income accounting, p. 33
national saving, p. 40
net international investment position, p. 40
official foreign exchange intervention, p. 50
official international reserves, p. 50
official settlements balance (or balance of payments), p. 51
private saving, p. 42

PROBLEMS

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1. We stated in this chapter that GNP accounts avoid double counting by including only the value of *final* goods and services sold on the market. Should the measure of imports used in the GNP accounts therefore be defined to include only imports of final goods and services from abroad? What about exports?
2. Equation (2-2) tells us that to reduce a current account deficit, a country must increase its private saving, reduce domestic investment, or cut its government budget deficit. Nowadays, some people recommend restrictions on imports from China (and other countries) to reduce the American current account deficit. How would higher U.S. tariffs on imports affect its private saving, domestic investment, and

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government deficit? Do you agree that import restrictions would necessarily reduce a U.S. current account deficit?

3. Explain how each of the following transactions generates two entries—a credit and a debit—in the Japanese balance of payments accounts, and describe how each entry would be classified:
   a. A Japanese investor buys a share of a Thailand stock company, paying by writing a check on an account with a Singapore bank.
   b. A Japanese investor buys a share of a Thailand stock company, paying the seller with a check on a Japanese bank.
   c. The Malaysian government carries out an official foreign exchange intervention in which it uses Yen held in a Japanese bank to buy ringgit (Malaysian currency) from its citizens
   d. A tourist from Kyoto buys a meal at an expensive restaurant in Bangkok, paying with a traveler's check.
   e. A saké brewery in Kobe contributes a case of saké bottles for a Paris wine-tasting event.
   f. A Japanese-owned factory in Germany uses local earnings to buy additional machinery.
4. A Mexican travels to Brazil to buy a gemstone which costs 3000 Real (Brazilian currency). The Brazilian company that sells the gemstone then deposits the 3000 Real in its account in a Panama bank. How would these transactions show up in the balance of payments accounts of Mexico and Brazil? What if the Mexican pays cash for the gemstone?
5. The nation of Pecunia had a current account deficit of $1 billion and a nonreserve financial account surplus of $500 million in 2017.
   a. What was the balance of payments of Pecunia in that year? What happened to the country's net foreign assets?
   b. Assume that foreign central banks neither buy nor sell Pecunian assets. How did the Pecunian central bank's foreign reserves change in 2017? How would this official intervention show up in the balance of payments accounts of Pecunia?
   c. How would your answer to (b) change if you learned that foreign central banks had purchased $600 million of Pecunian assets in 2017? How would these official purchases enter foreign balance of payments accounts?
   d. Draw up the Pecunian balance of payments accounts for 2017 under the assumption that the event described in (c) occurred in that year.
6. Can you think of reasons why a government might be concerned about a large current account deficit or surplus? Why might a government be concerned about its official settlements balance (that is, its balance of payments)?
7. Do the data on the European Central Bank, which set up the official settlements balance for the Eurozone, give an accurate picture of the extent to which foreign central banks buy and sell euros in currency markets?
8. Is it possible for a country to have a current account deficit at the same time it has a surplus in its balance of payments? Explain your answer, using hypothetical figures for the current and nonreserve financial accounts. Be sure to discuss the possible implications for official international reserve flows.
9. Suppose the South Africa net foreign debt is 50 percent of its GDP and foreign assets and liabilities pay an interest rate of 6 percent per year. What would be the drain on SA GDP (as a percentage) from paying interest on the net foreign debt? Do you think this is a large number? What if the net foreign debt were 100 percent of GDP? At what point do you think a country's government should become worried about the size of its foreign debt?

10. In 2015, in the balance of payment of Switzerland, the labor income in the primary income balance shows a deficit of 21 billion CHF (Swiss Franc), which represents almost 30 percent of the net account surplus (71 Billion CHF). How can you explain the size of this deficit?
11. Return to the example in this chapter's final Case Study of how a 10 percent dollar depreciation affects U.S. net foreign wealth (pp. 51–55). Show the size of the effect on *foreigners'* net foreign claims on the United States measured in dollars (as a percent of U.S. GDP).
12. We mentioned in the chapter that capital gains and losses on a country's net foreign assets are not included in the national income measure of the current account. How would economic statisticians have to modify the national income identity [equation (2-1)] if they wish to include such gains and losses as part of the definition of the current account? In your opinion, would this make sense? Why do you think this is not done in practice?
13. The international investment position of the European Union (28 countries) is negative of €2,557.4 billion in 2015, from €3,337.8 billion in 2014. The breakdown by country is as follows:

	Assets	Liabilities	Net	Net (% GDP)
Belgium	1,972.5	1,719.2	253.3	61.9
Bulgaria	37.9	64.7	−26.8	−60.7
Czech Republic	169.6	221.3	−51.7	−31.0
Denmark	793.8	682.1	111.7	42.0
Germany	7,900.6	6,418.6	1,482.0	49.0
Estonia	28.4	36.7	−8.3	−40.5
Ireland	4,747.4	5,279.5	−532.1	−208.0
Greece	243.4	477.5	−234.1	−133.0
Spain	1,640.9	2,615.8	−974.9	−90.2
France	6,422.1	6,780.2	−358.1	−16.4
Croatia	30.7	64.8	−34.1	−77.6
Italy	2,362.8	2,758.4	−395.6	−24.2
Cyprus	207.5	230.1	−22.5	−129.2
Latvia	30.5	44.9	−14.5	−59.3
Lithuania	22.9	39.7	−16.8	−45.2
Luxembourg	9,470.5	9,454.0	16.5	31.6
Hungary	269.7	339.5	−69.8	−64.1
Malta	223.1	219.1	4.0	45.4
Netherlands	7,583.2	7,171.8	411.4	60.8
Austria	892.9	877.4	15.5	4.6
Poland	209.7	471.2	−261.6	−61.1
Portugal	333.0	529.2	−196.2	−109.4
Romania	57.7	138.2	−80.6	−50.2
Slovenia	40.3	55.1	−14.8	−38.5
Slovakia	51.8	106.1	−54.3	−69.6
Finland	697.8	705.8	−8.0	−3.8
Sweden	1,299.1	1,305.7	−6.6	−1.5
United Kingdom	13,443.1	13,809.8	−366.6	−14.2

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What are the main creditor countries? What are the main debtor countries? Explain. (Hint: refer to the analysis on Eurostat analysis: http://ec.europa.eu/eurostat/statistics-explained/index.php/Economy_and_finance)

FURTHER READINGS

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